Assignment 3: Question 1 - Brand Loyalty

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Purpose and data

The aim is to compare households by their degree of brand loyalty, measured through the concentration of purchase volume. The analysis uses the three-household example to examine Euclidean distance, then constructs and checks `MaxBrShare` for the 600 households in `bathsoap.csv`.

Open `ETW2500_A3.Rproj` and render this document from a fresh R session. File paths are relative to the project root; package requirements and reproduction instructions are in `README.md`. All analysis code and outputs are shown below.

```
```{r}
library(tidyverse)
library(here)
library(gt)

brand_example <- here("dataset", "brand_example.csv") |>
  read_csv(show_col_types = FALSE)

BathSoap <- here("dataset", "bathsoap.csv") |>
  read_csv(show_col_types = FALSE)
```
```

Household	BrandA	BrandB
X	1.0	0.0
Y	0.0	1.0
Z	0.5	0.5

Part A: Think about the problem

1. Compare the loyalty of X and Y

Households X and Y are equally loyal in terms of their purchase shares: each allocates 100% of its purchase volume to one brand. X buys only brand A, while Y buys only brand B. Brand loyalty depends on how concentrated purchases are in a particular brand, rather than the identity of that brand. Household Z, which divides its purchases equally between two brands, is less loyal by this measure.

2. Euclidean distances using the brand shares

```
```{r}
brand_example |> gt()

distance_shares <- brand_example |>
  select(BrandA, BrandB) |>
  dist(method = "euclidean")

distance_shares
```
```

```
          1         2
2 1.4142136
3 0.7071068 0.7071068
```

The rows in the distance output correspond to X (1), Y (2), and Z (3). The household identifier is excluded because it is a label, not a measure of purchase behaviour.

Both variables are proportions on the same scale, so distances are calculated directly from the supplied shares. Euclidean distance compares the differences in the two coordinates; it does not identify equal loyalty to different brands.

For example, the distance between X and Y is $\sqrt{(1-0)^2+(0-1)^2} = \sqrt{2}$. X and Z are equally as close as Y and Z, while X and Y are the most distant pair. This does not agree with their similarity in loyalty in Item 1: X and Y are equally loyal, but the distance treats them as most different because they buy different brands.

3. Effect of using all nine brand-share variables unchanged

Using the nine brand shares unchanged makes distance-based methods compare which brands households buy and their shares. As a result, households that are equally loyal to different brands can be placed far apart, and clusters can reflect brand preference rather than degree of loyalty. Also, a high share of `BrOthers` can represent purchases spread across several small brands, so it cannot by itself establish loyalty to a single brand.

Part B: Create the variable

4. Create MaxBrShare for each household

First, check that each household has a unique ID and that the eight named brand shares have no missing values.

````
```{r}
BathSoap |>
  summarise(
    households = n(),
    unique_IDs = n_distinct(MemberID)
  )

BathSoap |>
  summarise(across(Br57_144:Br5, ~sum(is.na(.x))))
```
````

```
# A tibble: 1 x 2
  households unique_IDs
       <int>      <int>
1        600        600
# A tibble: 1 x 8
  Br57_144  Br55 Br272 Br286  Br24 Br481 Br352   Br5
     <int> <int> <int> <int> <int> <int> <int> <int>
1        0     0     0     0     0     0     0     0
```

There are 600 rows and 600 unique household IDs, with no missing values in the eight named brand-share variables. Therefore, grouping by `MemberID` gives one row per group. Within each group, `max()` compares the eight named brand shares for that household only.

For household i, the measure is $\text{MaxBrShare}_i = \max(s_{i1}, \ldots, s_{i8})$, where s_{ij} is its share of purchase volume for named brand j. Grouping is essential here: without it, `max()` would return one maximum across all households and assign that same value to every row.

```{r}
BathSoap <- BathSoap |>
  group_by(MemberID) |>
  mutate(
    MaxBrShare = max(
      Br57_144, Br55, Br272, Br286,
      Br24, Br481, Br352, Br5
    )
  ) |>
  ungroup()
```

This calculates a separate maximum for each household and retains all the original columns. `BrOthers` is deliberately excluded because it combines several brands. `ungroup()` removes the grouping before subsequent analysis.

5. Check MaxBrShare and investigate zero values

```{r}
summary(BathSoap$MaxBrShare)

BathSoap |>
  summarise(
    missing = sum(is.na(MaxBrShare)),
    outside_0_1 = sum(MaxBrShare < 0 | MaxBrShare > 1)
  )
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.0000  0.1336  0.2995  0.3712  0.5716  1.0000
# A tibble: 1 x 2
  missing outside_0_1
    <int>       <int>
1       0           0
```

The minimum is 0 and the maximum is 1. There are no missing values and no values outside this interval. The mean is approximately 0.3712 and the median is approximately 0.2995.

```{r}
BathSoap |>
  filter(MaxBrShare == 0) |>
  count(NoBrands, BrOthers) |>
  gt()
```

NoBrands	BrOthers	n
1	1	22

Household	BrandA	BrandB	MaxBrShare
X	1.0	0.0	1.0
Y	0.0	1.0	1.0
Z	0.5	0.5	0.5

There are **22 households with `MaxBrShare = 0`**. All 22 have `NoBrands = 1` and `BrOthers = 1`: each purchased only one brand, and that brand belongs to the pooled `BrOthers` category. Their shares for all eight named brands are zero, so excluding `BrOthers` produces a maximum of zero. These households are actually single-brand purchasers; their zero values should not be interpreted as an absence of brand loyalty.

6. Recalculate the example distances using MaxBrShare

Each household appears once in the example, so grouping by `Household` again gives a separate calculation for each row.

```
```{r}
brand_example <- brand_example |>
  group_by(Household) |>
  mutate(MaxBrShare = max(BrandA, BrandB)) |>
  ungroup()

brand_example |> gt()

distance_max <- brand_example |>
  select(MaxBrShare) |>
  dist(method = "euclidean")

distance_max
```
```

```
    1   2
2 0.0
3 0.5 0.5
```

X and Y both have `MaxBrShare = 1`, while Z has `MaxBrShare = 0.5`. Only `MaxBrShare` is used in the second distance calculation.

The comparison table below is generated directly from the two distance calculations. The entries of each distance object are ordered X-Y, X-Z, and Y-Z.

Table 1: Euclidean distances before and after changing the loyalty measure

Pair	Brand shares	MaxBrShare
X and Y	1.4142	0.0000
X and Z	0.7071	0.5000
Y and Z	0.7071	0.5000

````
```{r}
#| label: tbl-distance-comparison
#| tbl-cap: "Euclidean distances before and after changing the loyalty measure"
tibble(
  Pair = c("X and Y", "X and Z", "Y and Z"),
  `Brand shares` = distance_shares[1:3],
  MaxBrShare = distance_max[1:3]
) |>
  gt() |>
  fmt_number(columns = c(`Brand shares`, MaxBrShare), decimals = 4)
```
````

X and Y now have distance zero and are the most similar pair, consistent with their equal loyalty. Z is equally distant from both because only half of its purchase volume goes to either brand. This resolves the brand-identity problem in the example: loyalty to different brands receives the same value when the largest purchase share is the same. The limitation concerning `BrOthers` in Item 5 still applies to the full dataset.

`MaxBrShare` summarises concentration during the observed period; it does not describe the order of purchases or how often a household switches brands. Households with the same maximum can also distribute their remaining purchases differently, so this variable does not capture every aspect of loyalty.

7. Why interpret MaxBrShare together with NoBrands?

`MaxBrShare` measures concentration in the largest named brand, while `NoBrands` adds information about the breadth of brand purchasing that this maximum alone cannot show. Reading them together also prevents a household with `MaxBrShare = 0` and `NoBrands = 1` from being incorrectly labelled disloyal when its only brand is included in `BrOthers`.